

TIM 172B, Lecture # 8 (1/29/26); MOT-II
(SCM)

Agenda

1. Adaptive Forecasting Method 3 : Level- & Trend-
corrected exp.
smoothing

2. Adaptive Forecasting Method 4 : Level-Trend-and
seasonality-
corrected
exp. smoothing

3. HW# 4 has been published & is due
on Thursday, next week;

4. in time for doing the Take-home Midterm

(Adapt) → 5. Inventory Management

1. Level - and - Trend - corrected exponential smoothing
(Holt's method)

Assumption (of the method) : Data has only level, L , and trend, T
(i.e., seasonality in the data, if it exists, is ignored.)

Given: Historical data : $D_1, D_2, \dots, D_n$

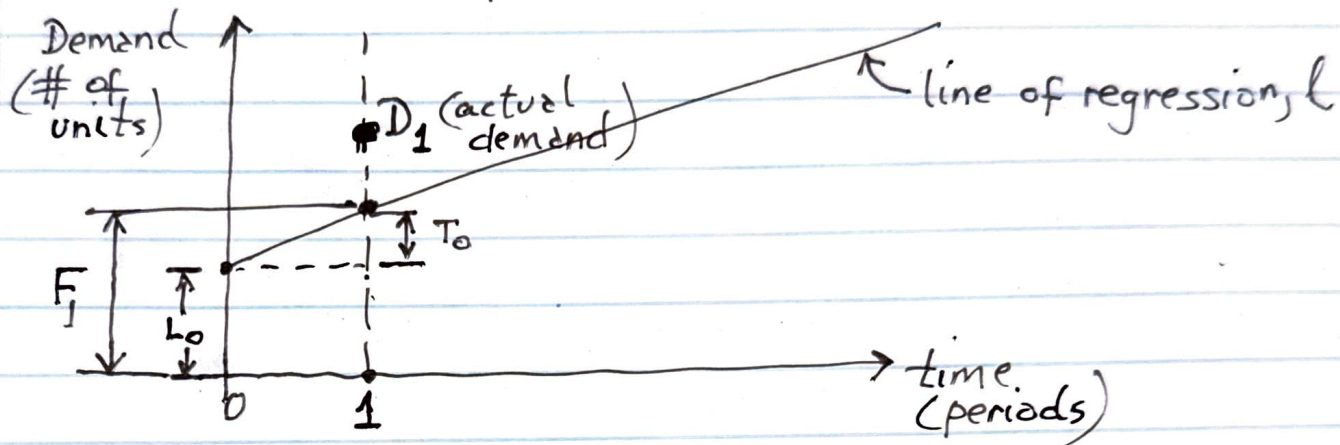
Determine: Forecasts : $F_1, F_2, \dots, F_n, \underbrace{F_{n+1}, F_{n+2}, \dots}_{\text{future}}$

Process:

Step 1 : Regress the given data using all available "n" data pts.; to compute initial values of level, L_0 , and trend T_0 .

Note : The data is not de-seasonalized before performing the linear regression.

Initial forecast of demand at time period 1 } , $F_1 = L_0 + T_0 \rightarrow (1)$



Actual demand at time $t_1 = D_1$

Step 2 : Adapt (or modify or correct)

Use 2 smoothing constants, α & β ,
 $(0 \leq \alpha \leq 1)$, $(0 \leq \beta \leq 1)$, to smooth level, L ,
 and the trend, respectively

for level, $L_1 = \alpha D_1 + (1-\alpha) [L_0 + T_0] \rightarrow (2)$

for trend, $T_1 = \beta (L_1 - L_0) + (1-\beta) T_0 \rightarrow (3)$

$F_2 = L_1 + T_1 \rightarrow (4)$
 (similar to (1))

Step 3 : Generalize (2), (3), and (4)

$$\text{Forecast, } F_{t+1} = L_t + T_t \rightarrow (5) \Rightarrow \boxed{\text{eqn. (14) in the textbook}}$$

Once we know the actual demand, D_{t+1} , at time $(t+1)$, then we can update level & trend, using

$$L_{t+1} = \alpha D_{t+1} + (1-\alpha) [L_t + T_t] \rightarrow (6) \Rightarrow \boxed{\text{eqn. (15) in the text}}$$

$$T_{t+1} = \beta [L_{t+1} - L_t] + (1-\beta) T_t \rightarrow (7) \Rightarrow \boxed{\text{eqn. 16 in the text}}$$

Example : Worked example in the text book
(HW #4, Prob.2)

1. Determine initial, L_0 & T_0 ("The seed")

Regress all the given 12 data pts. :
(8,000, ... - 41,000) to obtain
 $L_0 = 12,015$
 $T_0 = 1,549$

$$\text{Forecast, } \underset{(1)}{F_1} = L_0 + T_0 = 12,015 + 1,549 = 13,564$$

$$\text{actual demand, } D_1 = 8000$$

$$\text{error, } E_1 = 13,564 - 8,000 = 5,564$$

2. Adapt

$$\underset{(2)}{L_1} = \alpha D_1 + (1-\alpha)[L_0 + T_0] \rightarrow (a)$$

$$\text{Choose } \alpha = 0.1 \text{ \& } \beta = 0.2$$

$$L_1 = (0.1)(8000) + (0.9)(13,564) = 13,008$$

$$\underset{(3)}{T_1} = \beta(L_1 - L_0) + (1-\beta)T_0$$

$$= (0.2)[13,008 - 12,015] + (0.8)(1,549)$$

$$T_1 = 1,438$$

$$\bar{F}_2 = L_1 + T_1 = 13,008 + 1,438 = 14,446$$

actual demand, $D_2 = 13,000$

$$E_2 = \bar{F}_2 - D_2 = 1,446$$

Comments: The forecasts are getting better

Adaptive Forecasting, Method 4:

Level, Trend, and Seasonality corrected exponential smoothing (Winter's method)

Assumption (of the method): data has level, trend, and seasonality

1. Initialize the level, trend, and seasonal factors ("the seed"): $L_0, T_0, (S_1^0, S_2^0, \dots, S_p^0)$, using the static forecasting method.

$\Rightarrow$ the results of static forecasting provide the initial values of level, trend, and seasonality for Winter's method.

2. Initialize the forecast

$$F_1 = (L_0 + T_0) S_1^0$$

$$F_2 = (L_0 + 2T_0) S_2^0$$

...

...

$$F_p = (L_0 + pT_0) S_p^0$$

"p" forecasts
corresponding to
1 seasonal cycle.

3. Adapt (or modify or correct)

Use "three" smoothing constants,

$\alpha \rightarrow$ for smoothing level, $(0 \leq \alpha \leq 1)$

$\beta \rightarrow$ for smoothing trend, $(0 \leq \beta \leq 1)$

& $\gamma_i (i=1,2,\dots,p) \rightarrow$ for smoothing the
seasonal factors $(S_1, S_2, \dots, S_p)$,
 $(0 \leq \gamma_i \leq 1), (i=1,2,\dots,p)$,

to update the eqns. for level, trend, and
seasonality.

4. Forecasting [going from the present time, t ,
to the future, $(t+1)$]

1 cycle
of
forecasting

$$\left\{ \begin{array}{l} F_{t+1} = (L_t + T_t) S_{t+1} \\ \vdots \\ F_{t+p} = (L_t + pT_t) S_{t+p} \end{array} \right\} \begin{array}{l} \text{eqn. (17)} \\ \text{in the} \\ \text{textbook} \end{array}$$

5. Adapt

Once we know the actual demand at times $\underbrace{[(t+1), \dots, (t+p)]}_{1 \text{ cycle}} : D_{t+1}, D_{t+2}, \dots, D_{t+p},$

then the eqns. for updating level, trend, and the seasonal factors are given by:

EQN. for updating level : Egn. (18) in the textbook

EQN. for updating trend : Egn. (19) in the text

EQN. for updating seasonal factors : Egn. (20) in the text

$\Rightarrow$ Implement Winter's method, eqns. (17-20) of the text, in EXCEL

H.W. # 4, Prob. 2 : worked example in the text

Holt Winter

+
complete error analysis [9 error metrics]

Summary: Adaptive

The science:

all the theory
developed in
the lectures

The art:

Playing with the smoothing
constants, $(\alpha, \beta, \& \gamma_i)$
to adjust & improve the
forecast:



Structured playing:

For example:

Try low α → MAPE

Try high α → MAPE



Plot

↓
adjust

(TIM 1726 project)